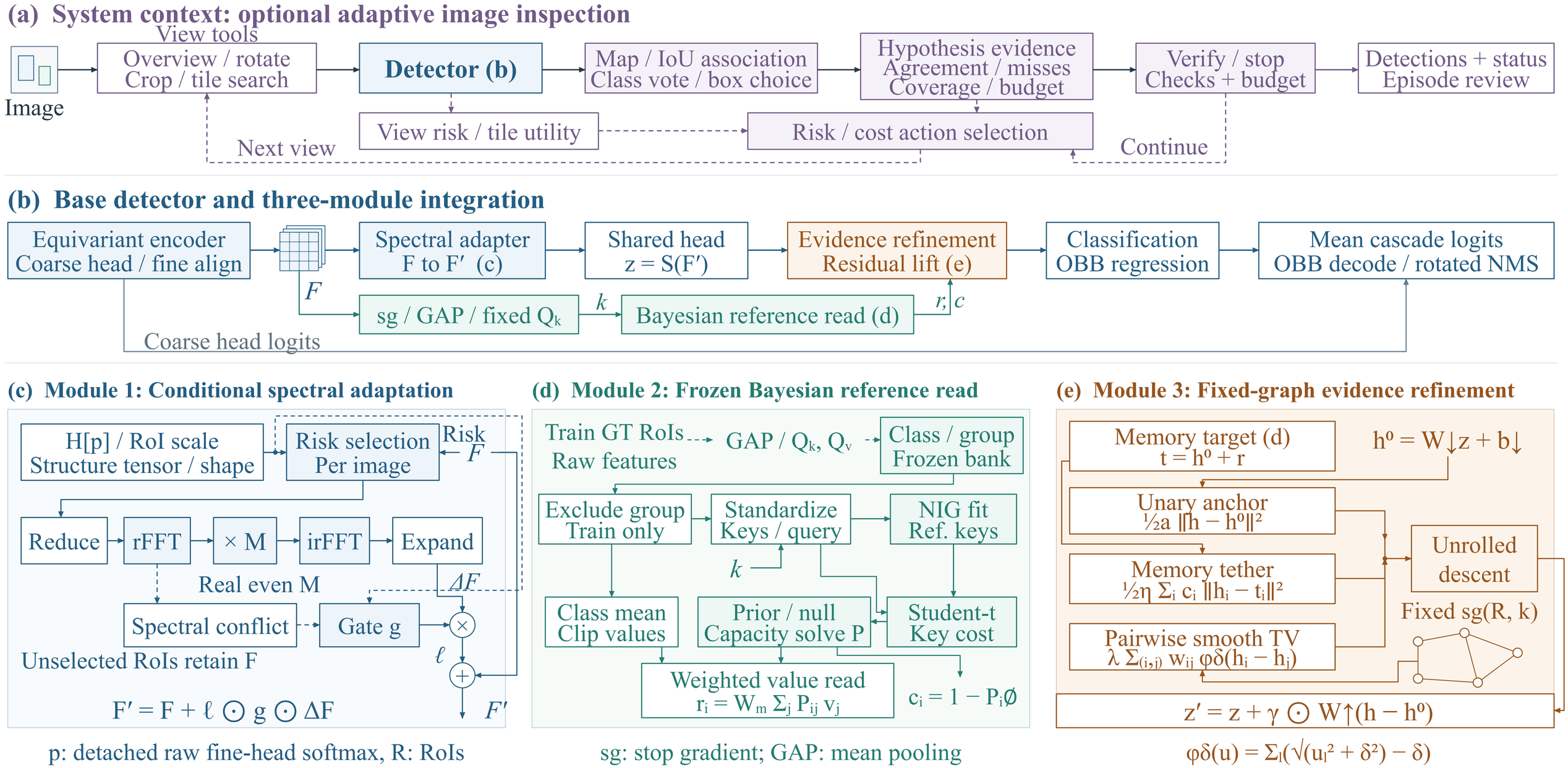

(a) System context: optional adaptive image inspection
View tools
Image
Overview / rotate
Crop / tile search
Detector (b)
Map / IoU association
Class vote / box choice
Hypothesis evidence
Agreement / misses
Coverage / budget
Verify / stop
Checks + budget
Detections + status
Episode review
View risk / tile utility
Risk / cost action selection
Next view
Continue
(b) Base detector and three-module integration
Equivariant encoder
Coarse head / fine align
F
Spectral adapter
F to F′ (c)
Shared head
z = S(F′)
Evidence refinement
Residual lift (e)
Classification
OBB regression
Mean cascade logits
OBB decode / rotated NMS
sg / GAP / fixed Qk
k
Bayesian reference read (d)
r, c
Coarse head logits
(c) Module 1: Conditional spectral adaptation
H[p] / RoI scale
Structure tensor / shape
Risk selection
Per image
Risk
F
Reduce
rFFT
× M
irFFT
Expand
ΔF
Real even M
Spectral conflict
Gate g
Unselected RoIs retain F
ℓ
F′ = F + ℓ ⊙ g ⊙ ΔF
F′
p: detached raw fine-head softmax, R: RoIs
(d) Module 2: Frozen Bayesian reference read
Train GT RoIs
Raw features
GAP / Qk, Qv
Class / group
Frozen bank
Exclude group
Train only
Standardize
Keys / query
NIG fit
Ref. keys
k
Class mean
Clip values
Prior / null
Capacity solve P
Student-t
Key cost
Weighted value read
ri = Wm Σj Pij vj
ci = 1 − Pi∅
sg: stop gradient; GAP: mean pooling
(e) Module 3: Fixed-graph evidence refinement
Memory target (d)
t = h⁰ + r
h⁰ = W↓z + b↓
Unary anchor
½a ‖h − h⁰‖²
Memory tether
½η Σi ci ‖hi − ti‖²
Pairwise smooth TV
λ Σ(i,j) wij φδ(hi − hj)
Unrolled
descent
Fixed sg(R, k)
z′ = z + γ ⊙ W↑(h − h⁰)
φδ(u) = Σl(√(ul² + δ²) − δ)